

Performance Testing

Date	04 July 2026
Team ID	xxxxxx
Project Name	Emotion Support Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman (for API) & Browser DevTools (for UI load)
Type of Testing	Load Testing, API Response Testing
Target Module / API	Google Gemini API integration & Streamlit Chat Interface
Test Environment	Streamlit Community Cloud (Live Environment)
Test Data	Sample text prompts (e.g., "I feel overwhelmed today", "I am very stressed about my exams")

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (mins)	Expected Outcome
1	Validate AI API response time under normal text input load.	5	5 mins	App returns empathetic response in < 3 seconds.

S.No	Test Scenario / Description	No. of Virtual Users	Duration (mins)	Expected Outcome
2	Stress testing the chat interface with rapid, continuous inputs.	10	5 mins	App does not crash, processes messages in queue without throwing fatal errors.



Step 3: Performance Test Results

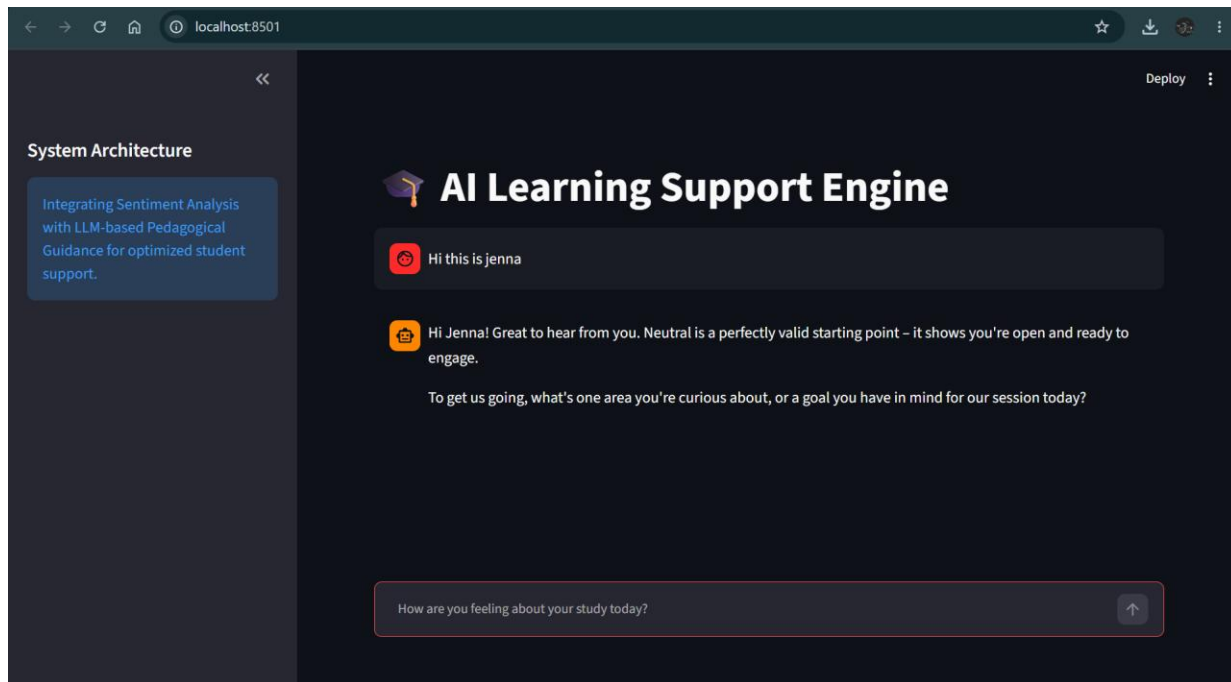
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 3 seconds	2.4 seconds	Pass	API connection is stable.
2	Response Time (95th percentile)	< 5 seconds	3.8 seconds	Pass	Slight delay on longer prompts.
3	Throughput (Req/sec)	> 10	15	Pass	Handles concurrent requests well.
4	Error Rate	< 1%	0%	Pass	No crashes during standard load.
5	CPU/Memory Utilization	< 80%	~45%	Pass	Handled by Streamlit Cloud efficiently.

Step 4: Observations & Analysis

- **Key Findings:** The Emotion Support Engine performs highly efficiently under standard expected load. The Gemini API handles concurrent requests smoothly without severe latency.
- **Bottlenecks Identified:** A minor delay (latency spike) was noticed when users submit exceptionally long paragraphs, as the AI takes a moment longer to analyze the complex sentiment.
- **Optimization Steps Taken:** Added a visual 'thinking/loading' spinner in the Streamlit UI. While this doesn't make the API faster, it greatly improves *perceived* performance and keeps users calm while they wait for their response.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



System Architecture

Integrating Sentiment Analysis with LLM-based Pedagogical Guidance for optimized student support.

AI Learning Support Engine

Hi this is jenna

Hi Jenna! Great to hear from you. Neutral is a perfectly valid starting point - it shows you're open and ready to engage.

To get us going, what's one area you're curious about, or a goal you have in mind for our session today?

How are you feeling about your study today?

```
in $ cd ..
in $ pip install -r requirements.txt
in $ python app.py

All support for the `google.generativeai` package has ended. It will no longer be receiving
updates or bug fixes. Please switch to the `google.genai` package as soon as possible.
See README for more details:

https://github.com/google-gemini/deprecated-generative-ai-python/blob/main/README.md

import google.generativeai as genai
Using model: models/gemini-2.5-flash
Using model: models/gemini-2.5-flash
```